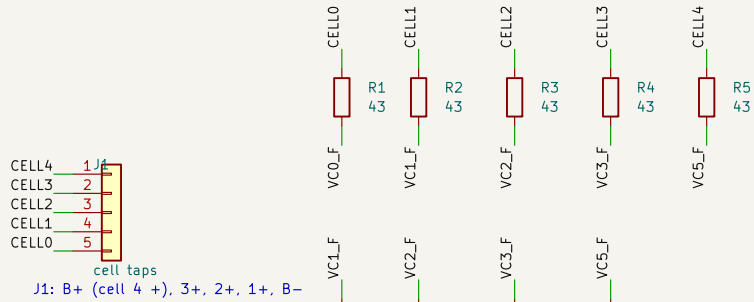
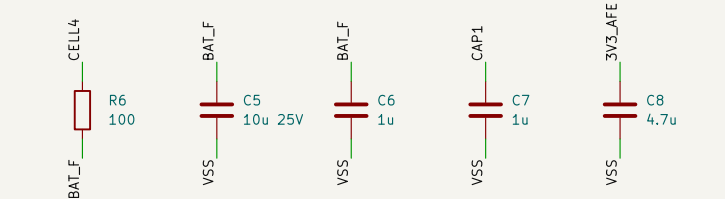


CELL TAPS + INPUT FILTERS



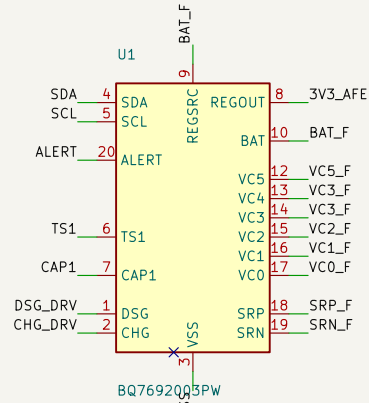
$R_c = 43\text{ ohm}$, $C_c = 1\text{ uF}$ per cell (datasheet range 40–1k ohm / 0.1–10 uF).
 R_c also sets the internal balancing current: $4.1\text{ V} / (2 \times 43 + -4)\text{ ohm} = 45\text{ mA}$.
4 cells on a BQ76920: VC4 is shorted to VC3 (datasheet Table 9–2).

SUPPLY



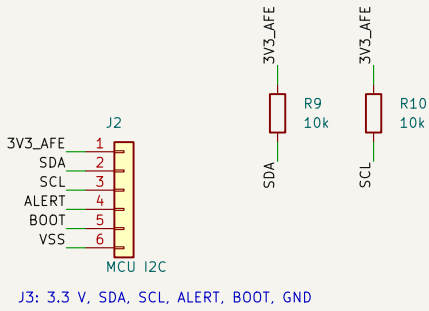
BAT filter $R_f = 100\text{ ohm}$ / $C_f = 10\text{ uF}$; REGSRC tied to BAT; REGOUT = 3.3 V for the MCU side

BQ76920 ANALOG FRONT END



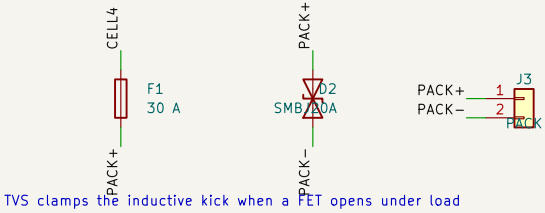
14-bit cell ADC (382 uV LSB), 16-bit coulomb counter (8.44 uV LSB),
hardware OV / UV / OCD / SCD protection, internal cell balancing

TO THE MCU (runs firmware/)



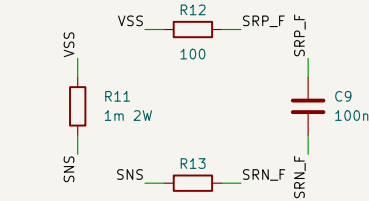
J3: 3.3 V, SDA, SCL, ALERT, BOOT, GND

PACK TERMINALS

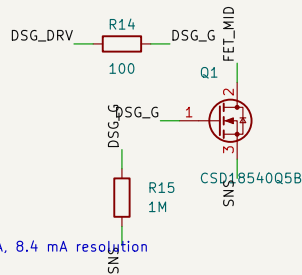


TVS clamps the inductive kick when a FET opens under load

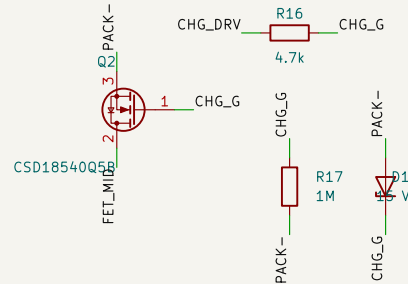
CURRENT SENSE + LOW-SIDE PROTECTION FETs



1 mOhm shunt: OCD 22.2 mV = 22 A, SCD 66.6 mV = 67 A, 8.4 mA resolution



Q1/Q2: CSD18540Q5B, 60 V, 1.8 mOhm (0.9 W at 22 A). Back-to-back FETs: DSG blocks discharge,
CHG blocks charge; each body diode lets the other direction flow, so a tripped pack can
still be charged after an under-voltage trip or discharged after an over-voltage trip.



Generated by hardware/gen_afe.py
Firmware: firmware/ (EKF SOC, protection, balancing) on the MCU at J3
4x LG HG2 18650 in series (16.8 V max), 22 A discharge / 4.5 A charge

Sheet: /		
File: bms_afe.kicad_sch		
Title: 4S Li-ion BMS – BQ76920 analog front end + protection FETs		
Size: A3	Date: 2026–10–07	Rev: A
KiCad E.D.A. 10.0.6		Id: 1/1